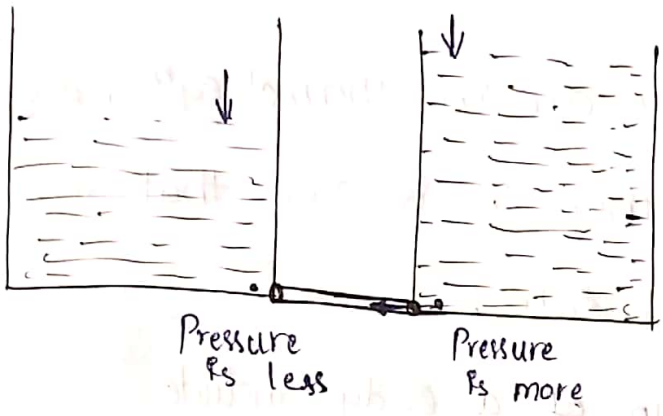
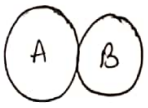
Heat & Thermodynamics

14/10/24 | Monday

Thermometry:

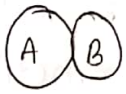


water flows from right vessel to left vessel.



$T_A > T_B$

due to their temp difference the heat transfered from hot body to cold body.



when $T_A = T_B$.

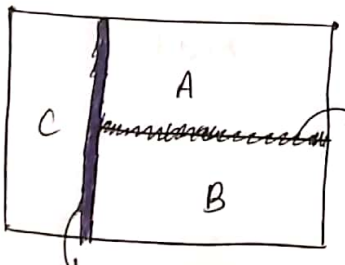
The bodies are said to be in thermal eqm.

the heat flow stops. Then

Note: If same heat is there for both bodies then it need not to be in thermal eqm.

Only thermal eqm is said when Temps are equal.

Zeroth law of thermodynamics:



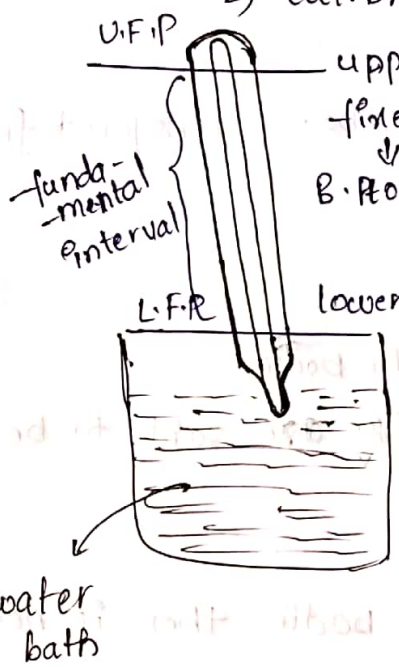
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If $T_A = T_C$ & $T_B = T_C$

then $\boxed{T_A = T_B}$

statements: If A & C are in thermal eqm, B & C are in thermal eqm, then it is said that A & B are also in thermal eqm.

* Measurement of temp of a body includes
  1) construction of an instrument
  2) calibration.

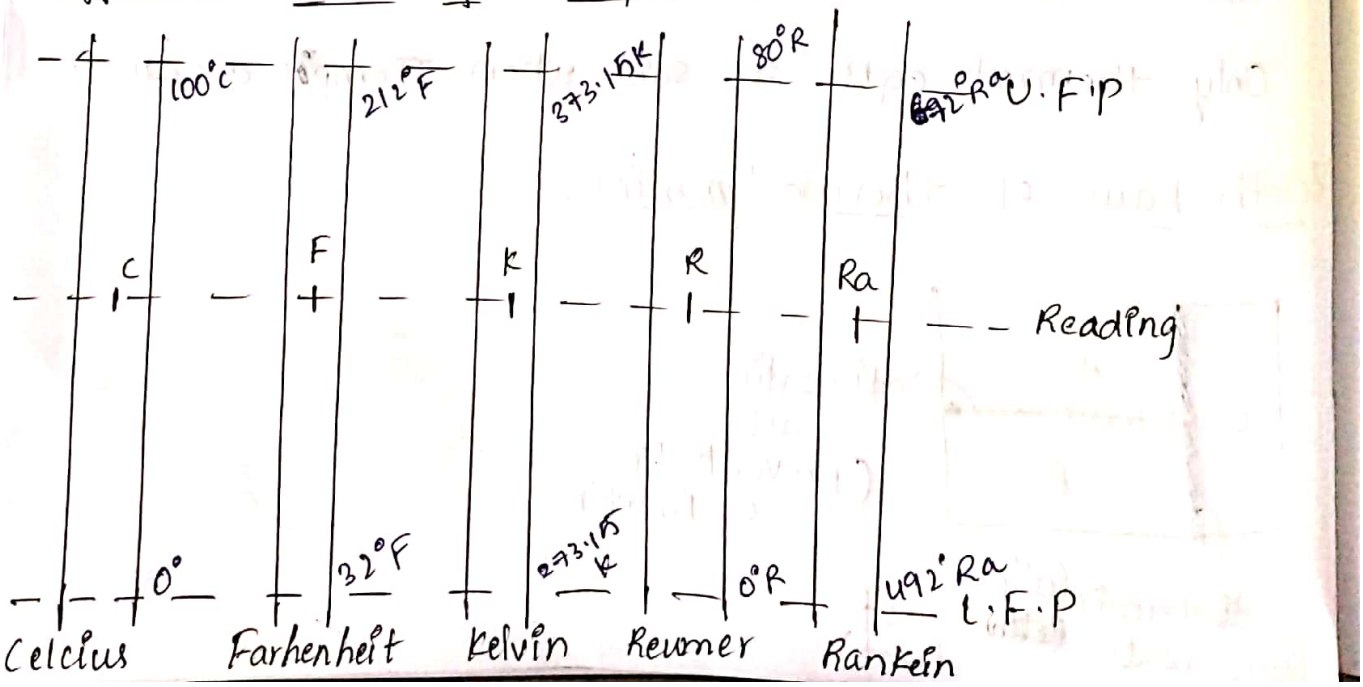


$\boxed{X = aT + b}$

(or) $\boxed{T = aX + b}$

* any property is linearly with temp → property of a thermometric substance changes

Differential Scales of Temperatures



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* A temp difference of

$$100°C = 180°F = 100K = 80°R = 180°Ra$$

* A temp difference of

$$1°C = 1.8°F = 1K = 0.8°R = 1.8°Ra$$

Relation b/w different scales of temp:

$$\frac{\text{Reading} - \text{L.F.P}}{\text{U.F.P} - \text{L.F.P}} = \text{const} \quad **$$

* also used for faulty-calibrated thermometer.

$$\Rightarrow \frac{C-0}{100-0} = \frac{F-32}{212-32} = \frac{K-273.15}{373.15-273.15} = \frac{R-0}{80-0} = \frac{Ra-492}{672-492}$$

Imp

$$\frac{C}{100} = \frac{F-32}{180} = \frac{K-273.15}{100} = \frac{R}{80} = \frac{Ra-492}{180} \quad ***$$

on differentiating,

$$\frac{\Delta C}{100} = \frac{\Delta F}{180} = \frac{\Delta K}{100} = \frac{\Delta R}{80} = \frac{\Delta Ra}{180}$$

$$\Rightarrow \Delta C = \Delta K \quad , \quad \Delta F = \Delta Ra$$

*points:

① Realation b/w C & K:

$$\frac{C}{100} = \frac{K-273.15}{100}$$

$$\boxed{C = K - 273.15} \quad \text{or} \quad K = C + 273.15$$

for ex,

$C = 1°C$

$K = 273.15 + 1$

$\boxed{K = 274.15K}$

$C = 10°C$

$\boxed{K = 283.15K}$

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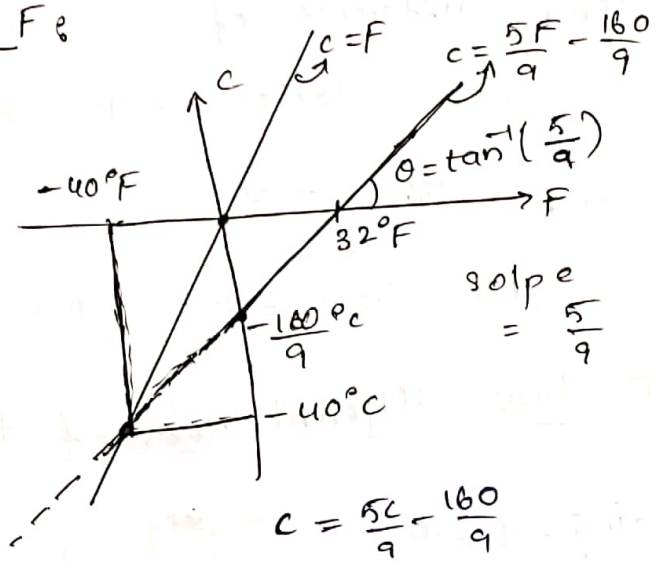
② Relation b/w C & F :

* $\frac{C}{100} = \frac{F-32}{180}$

$\frac{C}{5} = \frac{F-32}{9}$

$C = \frac{5F}{9} - \frac{160}{9}$

$y = mx + C$



$C = \frac{5C}{9} - \frac{160}{9}$

$\frac{4C}{9} = -\frac{160}{9}$

$C = -40°C$

for ex,

$F = 2C$

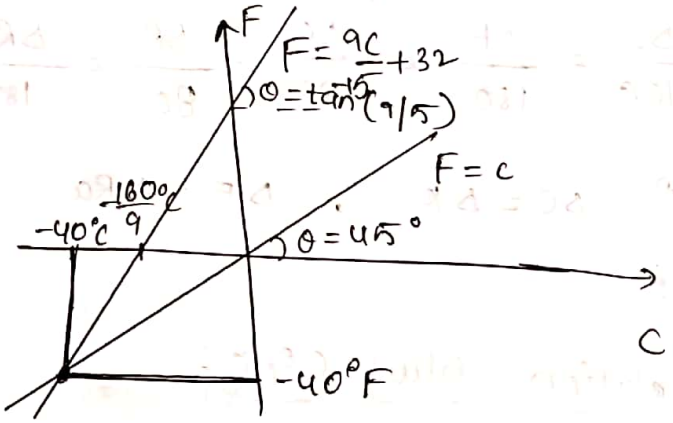
$C = \frac{10C}{9} - \frac{160}{9}$

$-\frac{C}{9} = -\frac{160}{9}$

$C = 160°C \Rightarrow F = 320°F$

* $\frac{9C}{5} + 32 = F$

$mx + C = y$



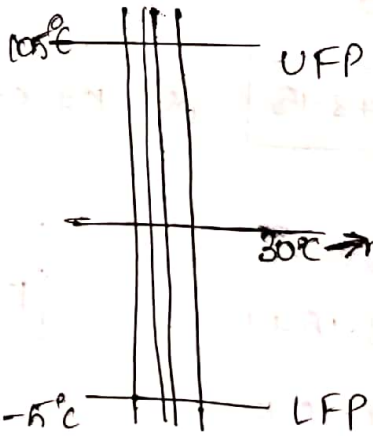
Faulty Thermometer :

$\frac{30-(-5)}{105-(-5)} = \frac{C-0}{100-0}$

$\frac{C}{100} = \frac{35}{110}$

$C = \frac{350}{11}$

$C = 31.8°C$ True value



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Error = measured value − true value.

$= 30 - 31.8 = -1.8^\circ C$

Correction = − Error = $1.8^\circ C$

Different types of Thermometers:

1) liquid Thermometer
2) Gas Thermometer
3) Thermo-emf thermometer
4) Platinum resistance thermometer
5) Radiation pyrometer.

we have,

$$X = at + b.$$

at $t = 0^\circ C \Rightarrow X = X_0$

$$X_0 = a(0) + b$$

$$X_0 = b$$

at $t = 100^\circ C \Rightarrow X = X_{100}$

$$X_{100} = a(100) + X_0$$

$$\frac{X_{100} - X_0}{100} = a$$

$$X_t = \frac{X_{100} - X_0}{100}(t) + X_0$$

$$\left(\frac{X_t - X_0}{X_{100} - X_0}\right) 100 = t \quad ***$$

X_t — property of substance at unknown temp.

X_0 — property of substance at $0^\circ C$.

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X_{100} = property of substance at $100^\circ C$.

$$\therefore * t = \left(\frac{l_t - l_0}{l_{100} - l_0}\right) \times 100$$

$$* t = \left(\frac{V_t - V_0}{V_{100} - V_0}\right) \times 100$$

$$* t = \left(\frac{P_t - P_0}{P_{100} - P_0}\right) \times 100$$

$$* t = \left(\frac{R_t - R_0}{R_{100} - R_0}\right) \times 100$$

Temperature coeff of Resistance (α) :-

we have,

$$R_t = R_0 (1 + \alpha t + \beta t^2 + \cdots)$$

neglecting higher order terms,

$$R_t = R_0 (1 + \alpha t) \quad **$$

$$\alpha = \frac{R_t - R_0}{R_0 (t)} \ /^\circ C$$

unit: $/^\circ C$ (or) $/K$ (or) $^\circ C^{-1}$ (or) K^{-1}.

(or)

$$\alpha = \frac{1}{R_0} \frac{dR_t}{dt}$$

* if $R_t \uparrow$ with increase in temp then $\alpha = +ve$.

* $\alpha = +ve$ — metals.

* $\alpha = -ve$ — semiconductors.

* if $R_t \downarrow$ with decrease in temp then $\alpha = +ve$.

now,

$$R_1 = R_0 (1 + \alpha t_1) \quad -①$$

$$R_2 = R_0 (1 + \alpha t_2) \quad -②$$

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then,

② - ① $R_2 - R_1 = R_0 \alpha (t_2 - t_1)$

$$\boxed{\alpha = \frac{R_2 - R_1}{R_0 (t_2 - t_1)}}$$

② / ① $\frac{R_2}{R_1} = \frac{1 + \alpha t_2}{1 + \alpha t_1}$

$R_2 + R_2 \alpha t_1 = R_1 + R_1 \alpha t_2$

$R_2 - R_1 = (R_2 t_1 + R_1 t_2)\alpha$

$$\boxed{\alpha = \frac{R_2 - R_1}{R_1 t_2 - R_2 t_1}} \quad ***$$

* In Modern Techniques :

* Triple pt of water is taken as fixed pt instead of 2 fixed pts *

$$\boxed{T_{tr} = 273.16\,K} \quad **$$

we have,

$T = ax + b$

at $T = 0\,K \Rightarrow X = 0$

$0 = a(0) + b$

$\boxed{b = 0}$

at $T = T_{tr} \Rightarrow X = X_{tr}$

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$$T_{tr} = a X_{tr}$$

$$\text{eq} \quad T = a X$$

$$\frac{T}{T_{tr}} = \frac{X}{X_{tr}}$$

$$T = T_{tr}\left(\frac{X}{X_{tr}}\right)$$

$$\Rightarrow \boxed{T = (273.16)\left(\frac{X}{X_{tr}}\right) K} \quad **$$

i.e,

$$* \quad T = 273.16\left(\frac{P}{P_{tr}}\right) K$$

$$* \quad T = 273.16\left(\frac{V}{V_{tr}}\right) K$$

Important pts :

*Liquid Thermometers :

* As mercury freezes at $-38.8^{\circ}C$ & boils at $356^{\circ}C$ the mercury thermometer cannot be used for very loco or high temp measurement.

*A mercury thermometer can be used to measure temp upto $300^{\circ}C$.

* Thermometer telling N_2 gas at high pressure above mercury column can be used to a temp upto $500^{\circ}C$.

* Alcohol thermometer is suitable for loco temp measurements upto $-180^{\circ}C$.

* Liquid pentane thermometer is used to measure V. loco temp upto $-200^{\circ}C$.

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* Gas Thermometers:

* A const Vol H-thermometer can be used to measure temp from -200°C to 500°C with platinum thermometric bulb. it can be used upto 1100°C with glased proclaim bulb.

* N_2 gas thermometers are used upto 1500°C

* He gas thermometers are used to measure low temp below -200°C.

* Const pressure air thermometer can be used to measure temp upto 600°C.

* Electrical Resistance thermometer can be used over a range of -272°C to 1200°C

* Saturated Vapour pressure of a liquid is used to measure very low temp below -268°C to -272°C

* Radiation thermometers are used to measure temp above 3000°C.

Q. Electrical resistance of lead wire varies with temp as $R = R_0 [1 + 3 \times 10^{-3} (T - T_0)]$. The resistance is 101.6Ω at the triple pt of water, 165.5Ω at normal (~~temperature~~) melting pt of lead (600.5K). what is the temp when resistance is 123.4Ω.

Given,

$$R = R_0 [1 + 3 \times 10^{-3} (T - T_0)]$$

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$$101.6 = R_0 [1 + 3\times10^{-3}(273 - T_0)] \quad -①$$

$$165.5 = R_0 [1 + 3\times10^{-3}(600.5 - T_0)] \quad -②$$

$$123.4 = R_0 [1 + 3\times10^{-3}(T - T_0)] \quad -③$$

Solving ① & ② we get R_0 & T_0

$$\frac{①}{②} \Rightarrow \frac{1016}{1655} = \frac{1 + 3\times10^{-3}(273 - T_0)}{1 + 3\times10^{-3}(600.5 - T_0)}$$

$$0.6138 + 0.6138 \times 3\times10^{-3}(600.5 - T_0) = 1 + 3\times10^{-3}(273 - T_0)$$

$$0.6138 + 1.8414\times10^{-3}\times(600.5 - T_0) = 1 + 3\times10^{-3}\times273 - 3\times10^{-3}\times T_0$$

$$(3\times10^{-3} - 1.8414\times10^{-3})T_0 = 819\times10^{-3} - 1105.8\times10^{-3} + 0.3862$$

$$10^{-3}(1.1586)\,T_0 = 0.0994$$

$$T_0 = \frac{0.0994}{1.1586}\times10^{3}$$

$$T_0 = 0.0858\times10^{3}$$

$$\boxed{T_0 = 85.8\,^{\circ}C}$$

put T_0 in ① & ②

$$\Rightarrow \boxed{R_0 = 65.061}$$

$$\frac{\frac{123.4}{65.061} - 1}{3\times10^{-3}} + 85.8 = T$$

$$\boxed{T = 384.697\,^{\circ}C}$$

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2. Calorimetry :-

units of heat :

① Calorie (C.G.S)

② Kilocalorie (M.K.S)

③ Joule (S.I)

calorie :

1gm water ⟶ temp has a range 14.5°C to 15.5°C at 1atm

$$\boxed{1\text{Kcal} = 1000\text{cal}}$$

$$\boxed{1\text{cal} = 4.186\ \text{J}}^{**}$$

Heat Supplied

- change the temp of the body
- Change the state of the body

Heat capacity :

The amount of heat required to raise the temp of whole body by 1° is called it's heat capacity.

$$\boxed{\text{Heat capacity} = \frac{dQ}{dT}}$$

$$dT - dQ$$
$$1 - \frac{dQ}{dT}$$

units :

S.I unit – J/K

C.G.S unit – cal/°C

Specific heat capacity (s) :

The amount of heat required to raise the temp of unit mass of the body through 1°C is called it's specific heat.

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$$\text{Specific heat} = \frac{1}{m}\cdot\frac{dQ}{dT} \quad **$$

$m \longrightarrow \frac{dQ}{dT}$

$1 \longrightarrow \frac{1}{m}\cdot\frac{dQ}{dT}$

* It depends on 1) nature of material of body 2) state of body.

units: S.I unit – J/kg-K

C.G.S unit – cal/g-°C

* $S_{water} = 1\ cal/gm-°C$

$= 1 \times 4.186\ J/10^{-3}\ kg\ K$

$= 4186\ J/kgK$

* $S_{ice} = 0.5\ cal/gm-°C$

* $S_{steam} = 0.5\ cal/gm-°C$ (vapour)

* $S_{hydrogen} = 3.5\ cal/gm-°C$ (highest of all)

* $S_{Radon} = 0.022\ cal/gm-°C$

* for a small change in temp, change of heat is large then that body has large specific heat.

Heat required to change the tempelature of body:

$$dQ = msdT \quad **$$

m – mass of body (Total)

S – Specific heat

dT – Change in temp

In general, * $Q = \int_{T_1}^{T_2} msdT = ms(T_2 - T_1)$ — applicable for all situations (for ex s is given as func of T)

$$Q = ms\Delta T \quad *$$

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we have,

$$s = \frac{1}{m} \cdot \frac{dQ}{dt}$$

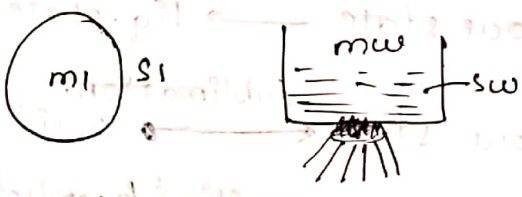
$$\boxed{\text{Heat capacity} = \frac{dQ}{dt} = ms} \quad **$$

* Heat capacity depends on 1) mass of body
2) specific heat of body

Water Equivalent of a body:

The mass of water in grams whose thermal capacity (heat capacity) is equal to thermal capacity of the given body.

Thermal capacity



$$= \frac{dQ}{dT} = m_1 s_1$$

Thermal capacity of water $= \frac{dQ_2}{dT_2} = m_w s_w$

By definition,

$$m_w s_w = m_1 s_1$$

$$\boxed{m_w = \frac{m_1 s_1}{s_w}}$$

$$s_w = 1 \text{ cal/gm-}^\circ\text{c}$$

$$\boxed{m_w = m_1 s_1} \quad ***$$

Different phases/states of Matter:

* Matter can exist in three different states (for physics)

1. Solids
2. Liquids
3. Gases.

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* solid state —melting→ liquid state
  (depends on pressure.)

* liq state —freezing→ solid state
  * also called solidification
  * pressure is const & temp is changing.

* liq state —boiling→ gaseous state

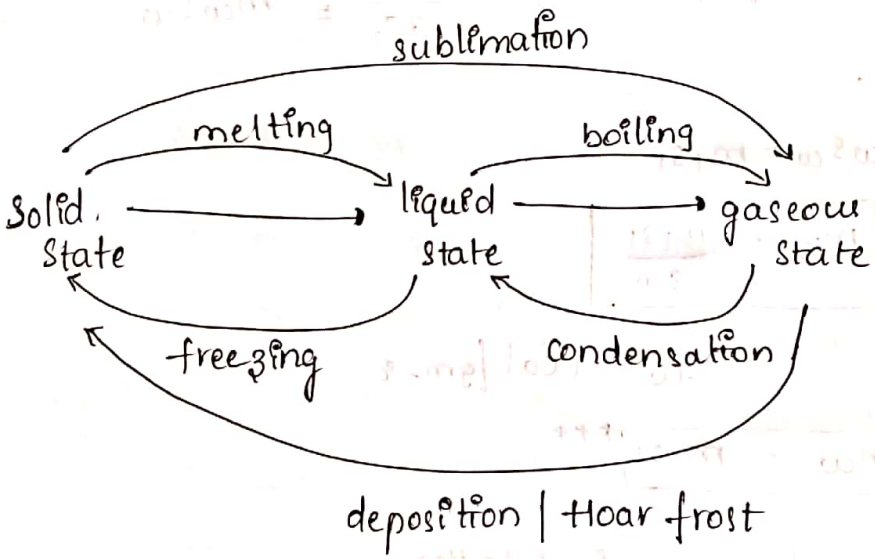
b.pt of water is 100°c at "normal atm pressure".

b.pt of water is 120°c at 'v.high pressure'.

* gaseous state —condensation→ liq. state

* gaseous state ←Sublimation— solid state

* Solid state ←deposition / Hoar frost— gaseous state



* for ex,
  ice is taken in a beaker at -20°c & started heating.
  1) change in temp from -20°c to 0°c.
  2) till total ice melt, there will be no change in temp.

∴ when temp change, State const & when state change, temp is

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Latent Heat:

The amount of heat required by unit mass of the body to change its state at const temperature is called latent heat.

$m \longrightarrow Q$

$1 \longrightarrow \frac{Q}{m}$

$$\boxed{\text{latent heat} = \frac{Q}{m}}$$

units: S.I – J/kg

C.G.S – cal/gm

Heat required to change the state of body at const temp:

$$\boxed{Q = mL}$$

m – mass of body that changes the state.

Latent heat of fusion of ice:

at const temp (0°C),

The amount of heat required by one gram of ice to change into water (at normal pressure) is called Latent heat of fusion of ice.

$$\boxed{L_{ice} = 80 \text{ cal/gm}}$$

$$= 80 \times 4.186 \text{ J}/10^{-3}\text{kg}$$

$$= 80 \times 4186 \text{ J/kg}$$

$$L_{ice} = 3.35 \times 10^{5} \text{ J/kg}$$

1gm ice ⟶ water ⟶ 80 cal.

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Latent of vapourisation of water:

at const temp (100°c),

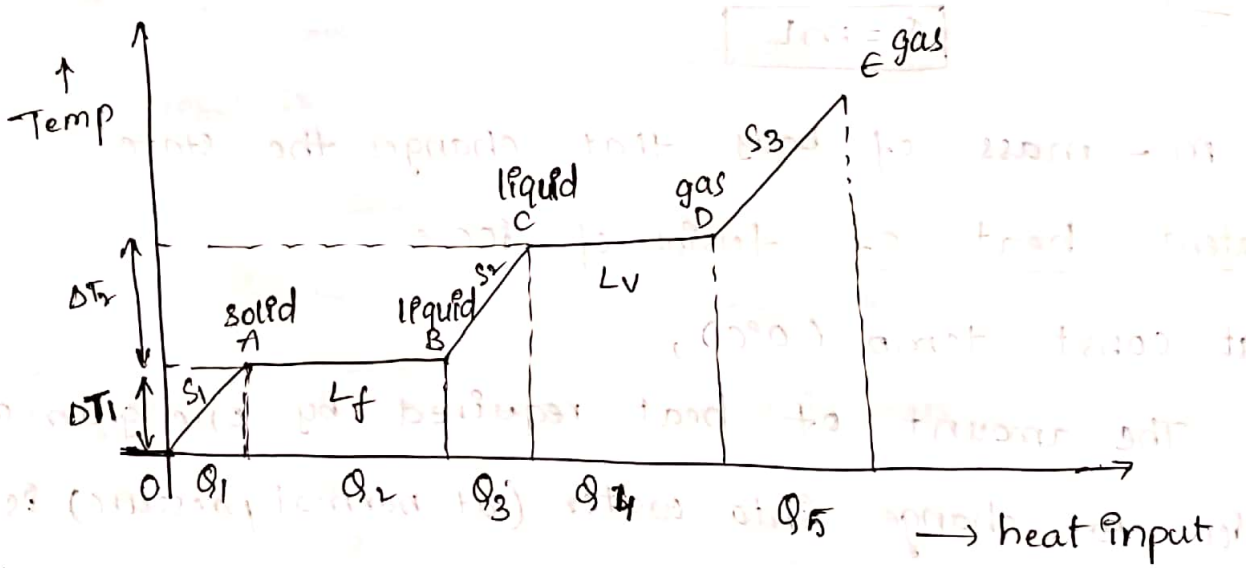
The amount of heat required to convert 1gm water into steam is called latent heat of steam.

$$L_{steam} = 540 \text{ cal/gm}$$

$$= 540 \times 4186 \text{ J/kg}$$

$$= 2.26 \times 10^6 \text{ J/kg}$$

Ex! Heat is supplied at a const rate to the solid. The temp of solid varies with heat input as shown.



from graph,

AB → melting

CD → boiling

If $CD = 2AB$

$$Q_4 = 2Q_2$$

$$mL_v = 2mL_f$$

$$\boxed{L_v = 2L_f}$$

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Again,

$$Q = ms\Delta T$$

$$\frac{\Delta T}{Q} = \frac{1}{ms}$$

$$\boxed{\text{Slope} = \frac{1}{ms}}$$

slope of line OA > slope of line BC.

$$\frac{1}{m_1 s_1} > \frac{1}{m_2 s_2}$$

$$\boxed{s_2 > s_1}$$

Principle of Mixtures

└ principle of calorimetry

when two or more non-reacting substances are placed in contact then heat lost by the hot body is equal to heat gained by cold body.

T_H = temp of hot body.

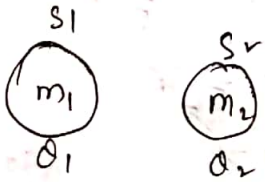
T_L = temp of cold body.

After mixing, T = resultant temp of all the contents. (common / final temp)

here we have,

$$\boxed{T_L \leq T \leq T_H}^{**}$$

Ex:



Here $\theta_1 > \theta_2$. when they are put in contact. Find resultant temp of the bodies.

θ = common temp.

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heat lost by hot body = heat gained by cold body.

$$m_1 s_1 (\theta_1 - \theta) = m_2 s_2 (\theta - \theta_2)$$

$$m_1 s_1 \theta_1 + m_2 s_2 \theta_2 = (m_2 s_2 + m_1 s_1)\theta$$

$$\theta = \frac{m_1 s_1 \theta_1 + m_2 s_2 \theta_2}{m_2 s_2 + m_1 s_1} \quad **$$

Ex: The temperature of equal masses of three different liquids A, B & C are 12°C, 19°C & 28°C respectively. The temperature when A & B are mixed is 16°C & when B & C are mixed is 23°C 1) what should be the temp when A & C are mixed. 2) what is the final temp if all the three liquids are mixed.

1) $\theta_{AB} = \frac{s_A \theta_A + s_B \theta_B}{s_A + s_B}$

$$16 s_A + 16 s_B = 12 s_A + 19 s_B$$

$$4 s_A = 3 s_B$$

$$\theta_{BC} = \frac{s_B \theta_B + s_C \theta_C}{s_B + s_C}$$

$$23 s_B + \frac{23}{5} s_C = 19 s_B + 28 s_C$$

$$4 s_B = \frac{5}{5} s_C$$

2) $\theta = \frac{s_A \theta_A + s_B \theta_B + s_C \theta_C}{s_A + s_B + s_C}$

$$= \frac{\frac{3}{4} s_B \times 12 + s_B \times 19 + 28 \times \frac{4}{5} s_B}{\left(\frac{3}{4} + 1 + \frac{4}{5}\right) s_B}$$

$$= \frac{9 + 19 + \frac{112}{5}}{\frac{15+20+16}{20}}$$

$$= \frac{\frac{28 \times 5 + 112}{5}}{\frac{51}{20}} = \frac{140 + 112}{51} = \frac{252}{51} = 19.76$$

$$\theta_{AC} = \frac{s_A \theta_A + s_C \theta_C}{s_A + s_C} = \frac{\frac{3}{4} s_B \times 12 + 28 \times \frac{4}{5} s_B}{\frac{3}{4} s_B + \frac{4}{5} s_B}$$

$$= \frac{45 + 112}{5\left(\frac{15+16}{20}\right)} = \frac{157 \times 4}{31} = 20.27°C$$

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Ex: One gram of steam at 100°C is passed over a insulating containel having one gram ice at 0°C. Find eqm temp of mixture. Neglecting the heat capacity of vessel.

Ice converts into water at 100°C

heat absorbed $\Rightarrow Q_1 = m L_{ice} + m s_w (100-0)$

$= 1(80) + (1)(1)(100)$

$Q_1 = 180$ cal

heat lost by steam that converts in to water at 100°C.

$\Rightarrow Q_2 = m L_{steam}$

$= (1)(540)$

$= 540$

clearly $\boxed{Q_1 \neq Q_2}$

∴ Mass of steam that condense into water at 100°C, $m_s = \frac{180}{540} = \frac{1}{3}$ gm

∴ final contents of mixture,

① mass of ice = 0

② mass of water = $1 + 1/3 = 4/3$

③ mass of steam = $1 - 1/3 = 2/3$

resultant temp = 100°C.

If m gms ice at 0°C is mixed with m gms of steam at 100°C then final mixture contains,

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1) water of mass $= \frac{4m}{3}$

2) steam of mass $= \frac{2m}{3}$

3) resultant mixture temp $= 100°C$.

→ 1) One gram ice at $0°C$ is placed in a calorimeter having one gram water at $40°C$. Find eqm temp & final contents. Assume heat capacity of calorimeter is negligibly small. $0°$, $1/2$ ice, $1/2$ water

2) one gram ice at $-40°C$ is placed in a calorimeter having one gram water at $10°C$. Find eqm temp & final contents. Assume heat capacity of calorimeter is negligibly small. $0°$, $\frac{9}{16}$ ice, $\frac{7}{8}$ water

3) In an expeirment on specific heat of a metal, A 0.2 kg block of the metal at $150°C$ is dropped in a copper calorimeter (of water equivalent 0.025 kg) containing $150\ cm^3$ of water at $27°C$. The final temp is $40°C$. Calculate the specific heat of the metal.

Sol:

3) heat lost by block $= ms\Delta T$

$$Q_1 = (0.2)(s)(150-40) \rightarrow (1)$$

heat gained by cold bodies,

$$Q_2 = m_c s_c \Delta T + m_w s_w \Delta T$$

$$= 0.025 \times 4186 \frac{J}{°C} (40-27)°C + 0.15\ kg \times 4186 \frac{J}{kg°C} (40-27)°C \rightarrow (2)$$

we have,

$$m_w = \frac{m_c s_c}{s_w}$$

$$0.025\ kg \times 4186 \frac{J}{kg°C} = m_c s_c$$

$$m_c s_c = 0.025 \times 4186 \frac{J}{°C}$$

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we have,

$$m_w = v_w s_w$$

$$= 150 \times 10^{-6}\ m^3 \times 10^3\ kg/m^3$$

$$m_w = 0.15\ kg$$

from principle of mixtures,

$$Q_1 = Q_2$$

$$0.2 \times S \times 110 = (0.025 + 0.15)\ 4186 \times 13$$

$$S = \frac{0.175 \times 4186 \times 13}{0.2 \times 110}$$

$$S = \frac{0.175 \times 2093 \times 13}{11 \times 2}$$

$$\boxed{S = 432.9}$$

1) One gm of ice at 0°C

↓

1 gm of water at 0°C

$$Q = mL = 80$$

1 gm water at 10°C

↓

1 gm of water at 0°C

$$Q = 1 \times 1 \times 40 = 40$$

$$m = \frac{40}{80} = \frac{1}{2}$$

$$m_i = 1/2 \qquad m_w = 1 + 1/2 = 3/2$$

2) 1 gm of ice at −40°C

↓

$$Q = 1 \times 0.5 \times 40 = 20$$

1 gm of ice at 0°C

↓

$$Q = 1 \times 80 = 80$$

1 gm of water at 0°C

$Q = 100$

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1gm of water at 10°c

↓

1gm of water at 0°c

$Q = 10$

$10 = m \times 80$

$m = 1/8$

$m_i = 1 + 1/8 = 9/8$

$m_w = 1 - 1/8 = 7/8$

Ex: Two spheres of radii in ratio 1:2 have specific heats in ratio 2:3. If their densities are in ratio 3:4. Find the ratio of their thermal capacities.

$$\frac{H_1}{H_2} = \frac{1}{8} \times \frac{2}{3} \times \frac{3}{4^2} = \frac{1}{16} \Rightarrow 1:16$$

Ex: A lead piece of mass 25gms gives out 1200 cal of heat. when it is cooled from 90°c to 10°c. what is it's 1) specific heat

2) heat capacity

3) water equivalent.

1) $S = \frac{1}{m} \cdot \frac{dQ}{dT}$

$$= \frac{1}{25} \times \frac{1200 \times 4.2}{80}$$

$$= \frac{\overset{3}{12} \times \overset{21}{42}}{25 \times \cancel{8}_2} = \frac{63}{25} \times \frac{4}{4} = \frac{252}{100} = 2.52 \times 1000$$

$$\boxed{S = 2520 \ \frac{J}{kg\text{-}c}}$$

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2) heat capacity

$$\frac{dQ}{dt} = ms = 63 \text{ J/K}$$

3) water equivalent

$$m_w = \frac{ms}{s_w} = \frac{63}{4200} \text{ kg} = 0.015 = 15 \text{ gms}.$$

Ex: A copper calorimeter of mass 300gms contains 500gms of water at a temp of 20°C. A 500gms of copper block at 100°C is dropped into the calorimeter. If the resultant temp is 25°C. Find specific heat of the copper.

$$25 = \frac{300 \times s \times 20 + 500 \times 20 \times 1 + 500 \times 100 \times s}{800s + 500}$$

$$25 = \frac{100 + 560s}{85 + 5}$$

$$25 = \frac{100 + 560s}{85 + 5}$$

$$2125 + 125 = 100 + 560s$$

$$2025 = 460s$$

$$s = \frac{25}{360}$$

$$s = \frac{5}{72} \times 4186$$

$$s = 290.7 \text{ J/kg-K}$$

$$25 = \frac{100 + 560s}{85 + 5}$$

$$85 + 5 = 4 + 80s$$

$$1 = 72s$$

$$s = \frac{1}{72} \times 4186$$

$$s = 290.7 \text{ J/kg-K}$$

$$\frac{5}{25} = \frac{100 + 560s}{85 + 5}$$

$$425 + 125 = 100 + 560s$$

Ex: A calorimeter of heat capacity 83.72 J/K contains 0.48kg of H_2O at 35°C. How much mass of ice at 0°C should be added to decrease temp of calorimeter to 20°C.

cold body — ice

hot body — calorimeter & water.

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heat lost by hot bodies

$$Q_1 = m_c s_c (25-20) + m_w s_w (25-20)$$

heat gained by cold bodies

$$Q_2 = m_i L_i + m_i s_w (20-0)$$

$$= m \times 80 \times 4186 + m \times 4186 \times 20$$

$$Q_1 = Q_2$$

$$\boxed{m = 0.07498 \text{ kg}}$$ ✓

Ex: A calorimeter of heat capacity 62.8 J/K contains 0.2kg of H_2O at 25°C. How much ~~water~~ mass of steam at 100°C should be added to increase the temp of calorimeter to 50°C.

heat lost by hot bodies,

$$Q_1 = m_s L_v + m_s s_w \Delta T$$

$$Q_1 = m_s (540 \times 4186 + 4186 \times 50)$$

heat gained by cold bodies,

$$Q_2 = m_c s_c \Delta T + m_w s_w \Delta T$$

$$Q_2 = (62.8 + 0.2 \times 4186) 25$$

$$Q_1 = Q_2$$

$$m_s (540 \times 4186 + 4186 \times 50) = (62.8 + 0.2 \times 4186) 25$$

$$m_s = \frac{950 \times 25}{590 \times 4186} \approx 9.1 \text{ gms}$$

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Ex: 20gms of ice at $-10^\circ C$ is converted into steam at $120^\circ C$. Find heat absorbed.

ice $-10^\circ C$ → ice $0^\circ C$ → water $0^\circ C$ → water $100^\circ C$ → steam $100^\circ C$ → steam $120^\circ C$

$$Q = m_i s_i (0-(-10)) + m_i L_i + m_i s_w (100-0) + m_i L_s + m_i s_v (20)$$

$$Q = m_i (10 s_i + L_i + 100 s_w + L_s + 20 s_v)$$

$$Q = 20(10 \times 0.5 + 80 + 100 \times 1 + 540 + 20 \times 0.5)$$

$$= 20(5 + 180 + 540 + 10)$$

$$= 20(735) = 14700 \text{ cal}$$

$$\boxed{Q = 14700 \text{ cal}}$$

Ex: 20gms of steam 100°c is passed into 100gms of ice at 0°c. Find resultant temp. 40°c

20gms of steam at 100°c ↓ $Q = 20 \times 540 = 10800$

200gms of water at 100°c

100gms of ice at 0°c, $Q = 8000$ ↓

100gms of water at 0°c, $Q = 10000$ ↓

100gms of water at 100°c

$2800 = 100 \times t \Rightarrow \boxed{t = 28^\circ C}$

$$T = \frac{20 \times 100 + 20 \times 100}{20 + 100} = \frac{2000 + 2800}{120} = \frac{4800}{120} = 40^\circ C$$

Ex: A calorimeter of water equivalent 10gms contains a liquid of mass 50gms at 40°c when m gms of ice at $-10^\circ C$ is put into the calorimeter the final temp was found to be 20°c. It is known that sp. heat capacity of liq changes with temp as $S = (1 + \frac{\theta}{500}) \frac{\text{cal}}{\text{gm}^\circ C}$ where θ = temp in °c. Find the value of m.

$m_w = 10 \text{gms}$

$\frac{m c s c}{s_w} = 10 \text{gms}$

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$$m_c s_c = 10\,gm \times \frac{1\,cal}{gm-{}^oC} = 10\,cal/{}^oC$$

heat gained by ice,

$$Q_1 = m_i s_i (0-(-10)) + m_i L_i + m_i s_w (20-0)$$

$$= 10 m_i s_i + m_i L_i + 20 m_i s_w$$

heat lost by calorimeter,

$$Q_2 = m_c s_c (40-20)$$

$$= 10 \times 20\,cal$$

$$= 200\,cal$$

heat lost by liquid,

$$Q_3 = \int_{20}^{40} m_l s \cdot d\theta$$

$$= m_l \int_{20}^{40} \left(1+\frac{\theta}{500}\right) \cdot d\theta$$

$$= m_l \left[\theta + \frac{\theta^2}{2(500)}\right]_{20}^{40}$$

$$= 50\,gms \left[20 + \frac{\cancel{20} \times \cancel{60}}{\cancel{2} \times \cancel{500}}\right]$$

$$= 50\,gms \left(\frac{100+6}{5}\right)$$

$$Q_3 = 1060\,cal$$

$$Q_1 = Q_2 + Q_3$$

$$m_i (10 \times 0.5 + 80 + 20 \times 1) = 1260$$

$$m_i (5 + 80 + 20) = 1260$$

$$m_i = \frac{1260}{105} = \frac{252}{21} = 12\,gms$$

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Ex:1 A refrigirator converts 1.3kg of H_2O at 20°c into ice at −15°c in 1hr. Calculate the effective power of refrigirator. given $L_f = 3.4 \times 10^5$ J/kg, $S_w = 4.2 \times 10^3$ J/kgk, $S_{ice} = 2.1 \times 10^3$ J/kg-k. P = 164.5 watt

Sol:

$$Q = m_w s_w (20) + m_w L_f + m_w s_{ice} (15)$$

$$= 1.3 (4.2 \times 20 \times 10^3 + 3.4 \times 10^5 + 2.1 \times 10^3 \times 15)$$

$$= 1.3 \times 10^3 (84 + 340 + 31.5)$$

$$Q = 592.15 \times 10^3$$

$$P = \frac{Q}{t} = \frac{592.15 \times 10^3}{60 \times 60} = \frac{5921.5}{36}$$

$$\boxed{P = 164.486 \text{ watt}}$$

Ex: How much steam at 100°c is to be passed into water of mass 100gms at 20°c to raise its temp by 5°c. 0.8139

$$m_s s_w \Delta T = m_s L_v + m_s s_w \Delta T$$

$$\frac{100}{1000} \times 4186 \times (25-20) = m_s \times 540 \times 4186 + m_s \times 4186 \times (100-25)$$

$$\frac{1}{2} \times 21 = m_s (540 + 75)$$

$$m_s = \frac{21}{2(615)} \text{ kg}$$

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$$m_s = \frac{500}{2(615)}$$

$$\boxed{m_s = 0.813 \text{ gms}}$$

Ex) A container contains small amount of water at 0°C. If air in the vessel is rapidly pumped out. it causes freezing of water. what percentage of water freezes out by this method?

Sol: At any temp there is some water vapour above surface of water the vapour pressure depends on temp. when air is pumped out the vapour is removed & vapour pressure falls. so there is some evaporation of liq to saturate the air.

The latent heat of vapourisation is supplied by water itself. So a part of it is freezes.

m = initial mass of water.

m_1 = mass of water that evaporates.

m_2 = mass of water that freezes.

we have,

$$m_2 L_f = m_1 L_v$$

$$m_2 = \frac{m_1 L_v}{L_f}$$

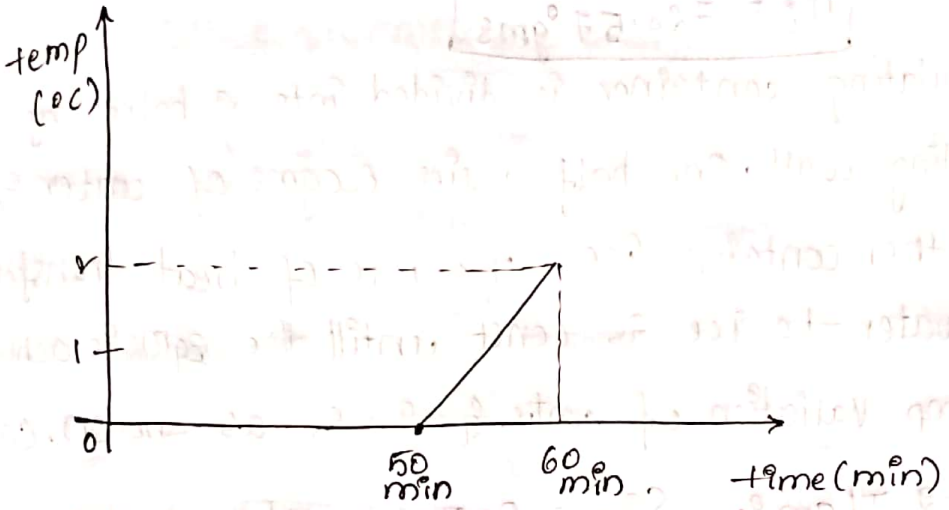
now,

$$\frac{m_2}{m} = \frac{m_2}{m_1 + m_2} = \frac{\frac{m_1 L_v}{L_f}}{m_1 + \frac{m_1 L_v}{L_f}} = \frac{L_v}{L_f + L_v}$$

$$\frac{m_2}{m} \times 100 = \frac{540}{620} \times 100 = \frac{27}{31} \times 100 = 87\%$$

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ex! A container contains 5kg of water at 0°c mixed with an unknown mass of ice in thermal eqm. water equivalent of container is 100gms. At time $t=0$, a heater is switched on which supplies heat at const rate to the container. The temp of mixture is measured at various times & is shown in the figure. Neglect heat lost from mixture container to surroundings. Calculate initial mass of ice.



sol: given power of heater is constant.

between 0 to 50 min,

$$P = \frac{dQ}{dt} = \frac{m_i L_f}{t}$$

$$= \frac{m_i (80\,cal/gm)}{50}$$

$$P = \frac{8}{5} m_i \quad -(1)$$

between 50 to 6 min

$$P = \frac{dQ}{dt} = \frac{m_c s_c \Delta\theta + m_w s_w \Delta\theta + m_i s_w \Delta\theta}{10}$$

$$P = \frac{100 \times 8 + 5000 \times 1 \times 2 + m_i \times 1 \times 1}{10 \times 5}$$

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$$P = \frac{5100 + m_i^\circ}{5} \quad -(2)$$

from eqn (1) & (2)

$$\frac{8m_i^\circ}{5} = \frac{5100 + m_i^\circ}{5}$$

$$7m_i^\circ = 5100$$

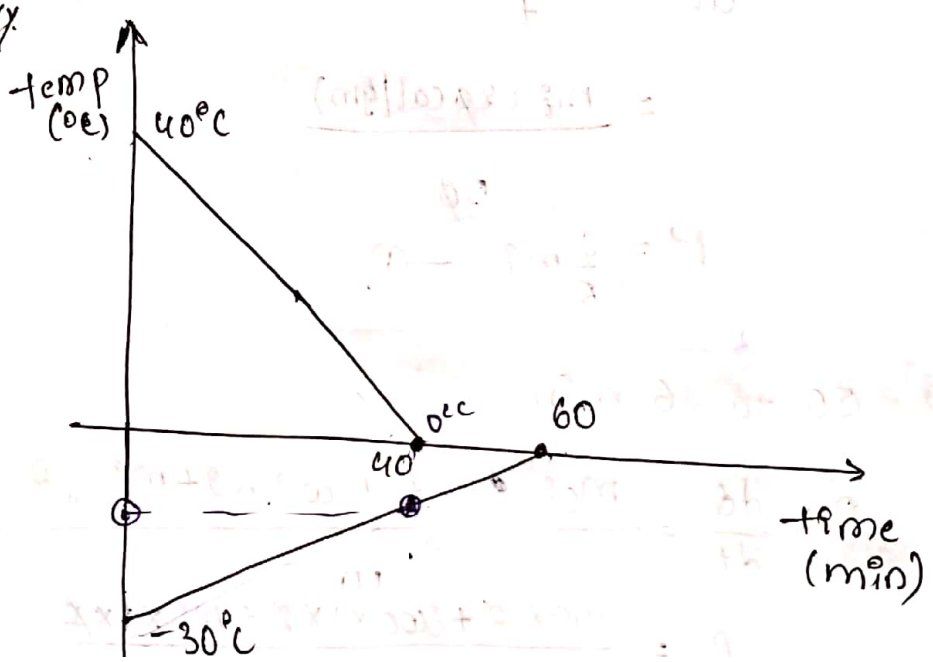
$$m_i^\circ = \frac{5100}{7}$$

$$\boxed{m_i^\circ = 728.57^\circ \text{ gms}}$$

Ex: A insulating container is divided into 2 halves by a conducting wall. One half carries 600gms of water & the other contains ice. The rate of heat transfer from water to ice is const until the eqm is achieved. The temp variation of water & ice is as shown. Given $S_w = 4.2$ J/gm°C, $S_{ice} = 0.5 \times 4.2$ J/gm°C & $L_{ice} = 80$ Cal/gm

Find 1) initial mass of ice in the container.

2) Mass of the water that converted into ice till thermal eqm is reached.



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Sol!! Given $P = \text{const}$

for water, $P = \frac{dQ}{dt} = \frac{m_w s_w (40-0)}{40} = m_w s_w$

$P = 600 \times 4.2 = 25.2 \times 100 = 2520$

for ice,

$$P = \frac{dQ}{dt} = \frac{m_i s_i (0-(-30))}{60}$$

$$P = \frac{m_i \times 2.1}{2}$$

$$2520 = \frac{m_i \times 2.1}{2}$$

$$600 \times 4.2 = \frac{m_i \times 2.1}{2}$$

$$\boxed{m_i = 2400 \text{ gms}}$$

time for water to freeze into ice = 60 - 40

$t = 20 \text{ min}$

we have,

$$P = \frac{Q}{t} = \frac{m_i L_i}{t} = \frac{m \times 80 \times 4.2}{20}$$

$$600 \times 4.2 = \frac{m \times 4 \times 4.2}{10}$$

$$m = 6 \times 25$$

$$\boxed{m = 150 \text{ gms}}$$

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Final contents:

① $m_{ice} = 2400 + 150 = 2550$ gms

② $m_w = 600 - 150 = 450$ gms

③ resultant temp $= 0°c$.

Ex: Ice at $0°c$ is added to 200gms of water initially at $70°c$ in a vaccum flask. when 50gms of ice has been added & has all melted & temp of the flask is $40°c$. when a further 80gms of ice is added & has all melted and the temp of whole becomes $10°c$. Find latent heat of fusion of ice in cal/gm.

let C = heat capacity of flask $= ms$

$$C(40-40) + m_w s_w (70-40) = 50L_i + 50s_w(40-0) \quad -①$$

and further,

$$C(40-10) + m_w s_w (40-10) = 80L_i + 80s_w(40-0) \quad -②$$

① $\Rightarrow$

$$30C + 200 \times 1 \times 30 = 50L_i + 50 \times 1 \times 40$$

$$30C + 6000 - 2000 = 50L_i$$

$$30C + 4000 = 50L_i$$

$$3C + 400 = 5L_i \quad -③$$

② $\Rightarrow$

$$30C + 250 \times 1 \times 30 = 80L_i + 80 \times 1 \times 10$$

$$3C + 750 = 8L_i + 80$$

$$3C + 670 = 8L_i \quad -④$$

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(4) - (3)

$$3L_f = 270$$

$$L_f = \frac{270}{3}$$

$$\boxed{L_f = 90 \text{ cal/gm}}$$

$$\frac{3C + 400}{5} = \frac{3C + 670}{8}$$

$$24C + 3200 = 15C + 3350$$

$$9C = 150$$

$$C = \frac{50}{3}$$

$$\boxed{C = 16.67 \text{ cal/}^\circ\text{C}}$$

Ex: A liq at 30°c is poured very slowly into calorimeter of 110°. The boiling temp of liq is 80°c. It is found that 1st 8gms of liq completely evaporates. after pouring another 100gms of liq the eqm temp was found to be 50°c. Find the ratio of latent heat of liq to it's specific heat.

Sol:

$$C(110-80) = 8S(80-30) + 8L$$

$$30C = 400S + 8L \quad -(1)$$

and further,

$$C(80-50) = 100S(50-30)$$

$$30C = 2000S \quad -(2)$$

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$$2000S = 400S + 8L$$

$$1600S = 8L$$

$$\boxed{200 = \frac{L}{S}}$$

Ex: A volume of 120ml of drink (1/2 alcohol + 1/2 water by mass) at a temp of 25°c is cooled by adding 20gm ice at 0°C. If all the ice melts. Find the final temp of drink. Given $S_{ice} = 0.833 g/cc$, $S_{alc} = 0.6$ cal/g°c.

Sol:

volume of 120ml of drink

$$V = 10 cm^3$$

mass of drink (m) $= V\rho$

$$= 120 \times 0.833$$

$$m = 100 gms$$

mass of alcohol, $m_1 = \frac{100}{2} = 50 gms$

$$m_w = m_2 = 50 gms.$$

$$m_{al} S_{al} (25-T) + m_w S_w (25-T) = m_{ice} L_f + m_i S_w (T-0)$$

$$50 \times 0.6 (25-T) + 50 \times 1 (25-T) = 20 \times 80 + 20 \times 1 \times T$$

$$(25-T)(50 \times 1.6) = 1600 + 20T$$

$$4 \times 80 (25-T) = 1600 + 20T$$

$$100 - 4T = 80 + T$$

$$20 = 5T$$

$$\boxed{T = 4°C}$$

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Effect of ~~melting pt~~ Pressure on melting pt :

"Classius - Clapeyron eqn"

$$\frac{dP}{dT} = \frac{L}{T(V_2 - V_1)}$$

V_1, V_2 → Specific volumes $\left(\frac{V}{m}\right)$ of solid & liq at absolute temp.

L = latent heat of fusion

T = absolute temp.

$\frac{dP}{dT}$ → ratio of change in pressure and change in melting pt.

points:

* Few substances like ice, iron, antimony etc, contracts on melting. For these substances melting pt decreases with increase of pressure & vise versa.
* Few substances like wax, copper etc., expands on melting. For these substances melting pt increases with increase in pressure.

Ex: The m.pt of ice is 0°C at 1atm. at what pressure will it be −1°C.

we have,

$$\frac{dP}{dT} = \frac{L}{T(V_2 - V_1)}$$

here $L = 80\ cal/gm = 80 \times 4186\ J/kg$

$T = 273\ K$, $P_1 = 1atm = 10^5\ N/m^2$

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$dT = -1 - 0 = -1\ K$

V_1 (ice) $= \frac{1}{0.9\ gm/cm^3} = \frac{1}{900\ kg/m^3}$

V_2 (water) $= \frac{1}{1000\ kg/m^3}$

now,

$$\frac{P_2 - P_1}{T_2 - T_1} = \frac{L}{T(V_2 - V_1)}$$

$$\frac{P_2 - 10^5}{-1} = \frac{80 \times 4186}{273\left(\frac{1}{1000} - \frac{1}{900}\right)}$$

$$P_2 - 10^5 = \frac{80 \times 4186}{273(100)} \times 1000 \times 900$$

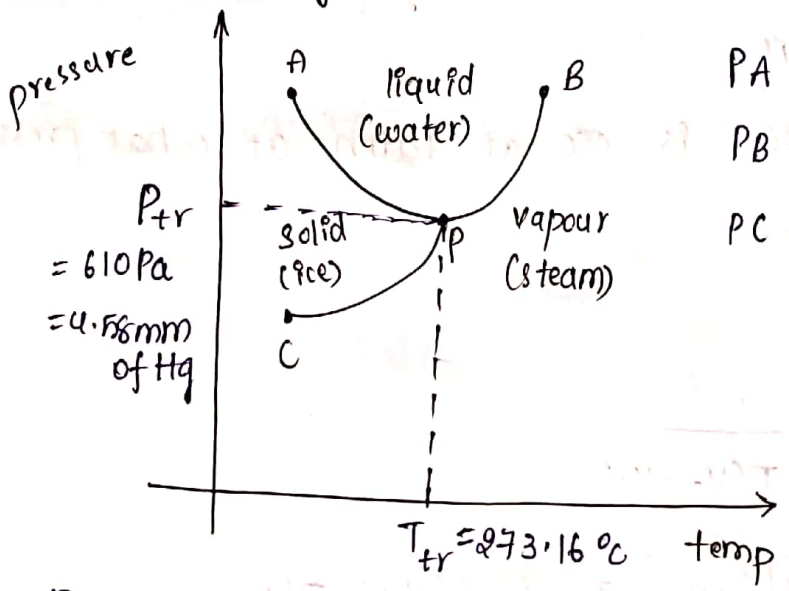
$$P_2 - 10^5 = \frac{8 \times 4186 \times 9}{273} \times 10^4 \times 10$$

$$P_2 = \left(\frac{8 \times 9 \times 418.6}{273} + 1\right) \times 10^5$$

$$= 111.4 \times 10^5\ Pa = 111.4\ atm$$

Triple point of water:

"phasar diagram"



PA ⟶ ice line

PB ⟶ steam line

PC ⟶ Sublimation line.

* Pressure ↑ ⇒ M.Pt ↓

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* pressure ↑ ⇒ B.Pt ↑

* pressure ↑ ⇒ Sub.pt ↑

* solid, liq, vapour coexist in eqm at P. Then this pt is called triple pt.

Joule's Law:

Quantity of heat produced is directly proportional to amount of work done.

$$W \propto Q$$

$$\boxed{W = JQ}$$

J — mechanical equivalent of heat or Joules constant.

$$\boxed{J = 4.2 \text{ J/cal}}.$$

$W \longrightarrow$ ① $W_f = -fs$

② $W_g = mgh$

③ $W_T = \Delta K.E$

$Q \longrightarrow$ ① $Q = ms\Delta T$

② $Q = mL$

Ex: A steel drilling machine making 180 rev/min is used to drill a hole in a block of steel. The mass of the steel block & drill is 180 gms. If the entire mechanical work is used up in producing heat & the rate of rise in temp of the block & drill is 0.5°C per second. Find ① rate of working of drill in watts. ② Torque required to drive the drill. Given specific heat of steel = 0.1 cal/g°C, J = 4.2 J/cal

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i)

rate of work done = $180 \times 0.1 \times 4.2 \times 0.5$

$= 18 \times \frac{1}{10} \times 4.2 \times \frac{1}{2}$

$= 9 \times 4.2$

$= 37.8 \frac{J}{sec}$

$\boxed{P = 37.8 \text{ watt}}$

ii) $P = \tau \omega$

$\tau = \frac{P}{\omega}$

$= \frac{37.8}{60 \times \frac{2\pi}{60}} = \frac{9 \times 4.2}{6\pi} = \frac{9 \times 0.7}{\pi} = \frac{6.3}{\pi}$

$= \frac{6.3}{3.14} = 2.005$

$\boxed{\tau = 2 N\text{-}m}$

Ex: A lead bullet is stopped by wooden obstacle. Assuming that 25% of heat is absorbed by obstacle. Find velocity of bullet if it's initial temp is 27°C. Given m.pt of lead = 327°C. $S_{lead} = 0.03 cal/g°c$ $(L_f)_{lead} = 6 cal/g$. Take $J = 4.2 J/cal$

$\frac{75}{100} \times \frac{1}{2} m v^2 = m(S\Delta T + L)$

$\frac{3}{8} v^2 = (0.03 \times 300 + 6) \times 4.2 \times 1000$

$v^2 = (9+6) 1.4 \times 8 \times 1000$

$= 15 \times 1.4 \times 8 \times 1000$

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$$V^2 = 12 \times 14 \times 1000$$

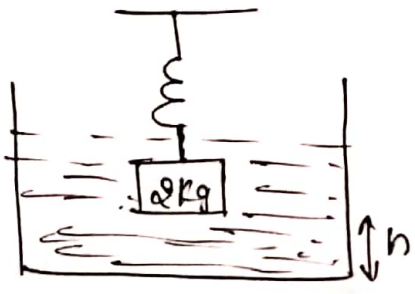
$$V = \sqrt{3 \times 4 \times 2 \times 7 \times 1000}$$

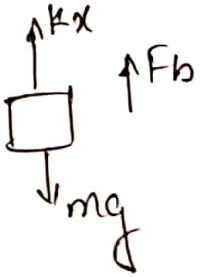
$$V = 20\sqrt{420}$$

$$\boxed{V = 409.87 \text{ m/s}}$$

Ex! A metal block of mass 8kg ($\rho = 5 \times 10^3 kg/m^3$) is suspended from a ideal spring of force const 25 N/m. The spring block system is immersed in 500gm of water contained in a vessel such that the block is at a height h from bottom of vessel The system is released so as to sink to the bottom resulting in a rise of temp of $0.004^\circ C$ of H_2O. The height h through which the block went down is 0.26m. Given specific heat of block = 250 J/kgK, Specific heat of water = 4200 J/kgK, heat capacity of vessel & spring are neglible. take $g = 10 m/s^2$.

Sol!





$$mg = Kx + F_b$$

$$80 = 25x + \frac{m}{\rho_b}\rho_l g$$

$$80 = 25x + \frac{80}{5}$$

$$25x = 16 \Rightarrow \boxed{x = 16/25 = 0.64m}$$

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$$mgh + \frac{1}{2}kx^2 = m_w s_w \Delta T + m_b s_b \Delta T$$

$$20h + \frac{1}{2} \times 25 \times \frac{256}{625} = (0.5 \times 4200 + 2 \times 250)(0.004)$$

$$20h + \frac{128}{25} = (2100 + 500)(0.004)$$

$$20h = 2600 \times 0.004 - \frac{128}{25}$$

$$= 10.4 - 5.12$$

$$20h = 5.28$$

$$h = \frac{0.528}{2}$$

$$\boxed{h = 0.264\,m}$$

Scanned with CamScanner